61. The element with positive electron gain enthalpy is

NCERT Page 90 / N-90

(a) hydrogen

(b) sodium

(c) oxygen

(d) neon

Answer: (d) neon

Explanation: Neon has a completely filled, very stable $2s^2 2p^6$ shell. Addition of another electron is energetically unfavourable, so its electron-gain enthalpy is positive.

62. Consider the following statements:
(i) The radius of an anion is larger than that of the parent atom.
(ii) The ionisation energy generally increases with increasing atomic number in a period.
(iii) The electronegativity of elements increases on moving down a group.
Which of the above statements is/are correct?

NCERT Pages 87 & 91 / N-87 & 91

(a) (i) alone

(b) (ii) alone

(c) (i) and (ii)

(d) (ii) and (iii)

Answer: (c) (i) and (ii)

Explanation: Anions are larger than their parent atoms, so (i) is correct. Ionisation energy generally rises across a period as effective nuclear charge increases, so (ii) is also correct. Electronegativity decreases, not increases, down a group, making (iii) false.

63. Which of the following metals requires radiation of the highest frequency to cause the emission of electrons?

NCERT Page 88 / N-88

(a) Na

(b) Mg

(c) K

(d) Ca

Answer: (b) Mg

Explanation: Photoelectric emission requires a threshold frequency proportional to the work function. Among Na, Mg, K and Ca, Mg holds its valence electrons most strongly and therefore requires the highest-frequency radiation.

64. Which of the following represents the correct order of increasing electron gain enthalpy with negative sign for the elements O, S, F and Cl?

NCERT Pages 89–90 / N-89–90

(a) Cl < F < O < S

(b) O < S < F < Cl

(c) F < S < O < Cl

(d) S < O < Cl < F

Answer: (b) O < S < F < Cl

Explanation: Considering increasing magnitude of negative electron-gain enthalpy: O is lower than S because the small 2p shell of O has greater electron repulsion; similarly F is lower than Cl. Halogens have larger electron-gain tendency than chalcogens, giving O < S < F < Cl.

65. An element A of group 1 shows similarity to an element B belonging to group 2. If A has maximum hydration enthalpy in group 1, then B is

NCERT Page 93 / N-93

(a) Mg

(b) Be

(c) Ca

(d) Sr

Answer: (a) Mg

Explanation: Li has the maximum hydration enthalpy among group 1 because Li^+ is the smallest alkali-metal ion. Lithium shows a diagonal relationship with magnesium, so the corresponding group 2 element B is Mg.

66. Which is the correct order of electronegativity?

NCERT Pages 90–91 / N-90–91

(a) F > O > N > C

(b) F > N > O > C

(c) C < N < O < F

(d) C > N > O > F

Answer: (a) F > O > N > C

Explanation: Electronegativity increases across period 2 from C to F as effective nuclear charge increases and atomic radius decreases. Hence F > O > N > C. Note: option (c), written C < N < O < F, expresses the same mathematical order; the question therefore contains a duplicated-equivalence issue.

67. The correct order of decreasing electronegativity values among the elements I—beryllium, II—oxygen, III—nitrogen and IV—magnesium is

NCERT Page 91 / N-91

(a) II > III > I > IV

(b) III > IV > II > I

(c) I > II > III > IV

(d) I > II > IV > III

Answer: (a) II > III > I > IV

Explanation: Using Pauling electronegativities approximately: O (3.44) > N (3.04) > Be (1.57) > Mg (1.31). Thus II > III > I > IV.

68. Which of the following elements is considered a metalloid?

NCERT Page 85 / N-85

(a) Sc

(b) Pb

(c) Bi

(d) Te

Answer: (d) Te

Explanation: Tellurium lies on the metalloid staircase and shows properties intermediate between metals and non-metals. Sc, Pb and Bi are metals.

69. The elements with zero electron affinity are

NCERT Pages 89–90 / N-89–90

(a) Boron and Carbon

(b) Beryllium and Helium

(c) Lithium and Sodium

(d) Fluorine and Chlorine

Answer: (b) Beryllium and Helium

Explanation: Be has a stable filled $2s^2$ subshell and He has the completely filled $1s^2$ shell. Electron addition is not favourable for either, so their electron affinity is effectively zero/very small in the simplified periodic-trend treatment.

70. Which is not the correct order for the stated property?

NCERT Pages 87–90 / N-87–90

(a) Ba > Sr > Mg; atomic radius

(b) F > O > N; first ionisation enthalpy

(c) Cl > F > I; electron affinity

(d) O > Se > Te; electronegativity

Answer: (b) F > O > N; first ionisation enthalpy

Explanation: The stated order is wrong because nitrogen has a stable half-filled $2p^3$ configuration. Therefore the correct relative order is F > N > O, not F > O > N.

71. The compounds of the s-block elements, with the exception of lithium and ...X..., are predominantly ionic. Here, X refers to

(a) hydrogen

(b) helium

(c) magnesium

(d) beryllium

Answer: (d) beryllium

Explanation: Most s-block compounds are ionic, but compounds of lithium and beryllium show considerable covalent character because the very small Li^+ and Be^{2+} ions have high polarising power.

72. Alkali metals are powerful reducing agents because

(a) these are metals

(b) their ionic radii are large

(c) these are monovalent

(d) their ionisation potential is low

Answer: (d) their ionisation potential is low

Explanation: Alkali metals lose their single valence electron very easily because of their low ionisation enthalpy. They are therefore readily oxidised and act as strong reducing agents.

73. An element having electronic configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$ forms

NCERT Page 94 / N-94

(a) acidic oxide

(b) basic oxide

(c) amphoteric oxide

(d) neutral oxide

Answer: (b) basic oxide

Explanation: The configuration is $[Ar]4s^1$, which is potassium, a group 1 metal. Potassium forms K_2O, a strongly basic oxide.

74. Which of the following sequences correctly represents the decreasing acidic nature of oxides?

NCERT Page 94 / N-94

(a) $Li_2O > BeO > B_2O_3 > CO_2 > N_2O_3$

(b) $N_2O_3 > CO_2 > B_2O_3 > BeO > Li_2O$

(c) $CO_2 > N_2O_3 > B_2O_3 > BeO > Li_2O$

(d) $B_2O_3 > CO_2 > N_2O_3 > Li_2O > BeO$

Answer: (b) $N_2O_3 > CO_2 > B_2O_3 > BeO > Li_2O$

Explanation: Across period 2, metallic character decreases and non-metallic character increases, so oxides become progressively more acidic. Therefore acidity decreases in the reverse order: $N_2O_3 > CO_2 > B_2O_3 > BeO > Li_2O$.

75. Which property of an element is directly related to electronegativity?

NCERT Pages 90–91 / N-90–91

(a) Atomic radius

(b) Ionisation enthalpy

(c) Non-metallic character

(d) None of these

Answer: (c) Non-metallic character

Explanation: Electronegativity measures the tendency of an atom in a bond to attract shared electrons. Greater electronegativity is associated with stronger non-metallic character, so the two are directly related.